3.3 & 3.4 Detecting Ranges with Branches

Java Programming Concepts

Learning Objectives

- Detect ranges using if-else-if statements
- Understand relational operators
- Apply range checking in Java programs
- Interpret sequential branching logic

Detecting Ranges Using if-else-if

- A common task is determining which range a value belongs to.
- Each condition only checks the upper bound because previous conditions are already false.
- This makes range detection simple and efficient.

Relational Operators

- < Less than
- > Greater than
- <= Less than or equal to
- >= Greater than or equal to
- != Not equal to

- Example: If x=3, then x<4 is true.

Insurance Price Example

- Age < 16 -> Too young
- Age 16-24 -> $4800
- Age 25-39 -> $2350
- Age 40+ -> $2100

- Sequential conditions identify the correct age range.

Java Range Detection Logic

- if (userAge < 16)

  Too young
- else if (userAge < 25)

  Age 16-24
- else if (userAge < 40)

  Age 25-39
- else

  Age 40+

- **Age < 16 -> Too young**
- **Age 16-24 -> $4800**
- **Age 25-39 -> $2350**
- **Age 40+ -> $2100**

- **Sequential conditions identify the correct age range.**

Summary

- • Relational operators compare values
- • if-else-if structures detect ranges
- • Sequential evaluation makes lower bounds implicit
- • Commonly used in decision-making programs